LUIS DEL ORBE

AI Software Engineer · Python, AWS & Generative AI

Santo Domingo, Dominican Republic · Remote or on-site · Open to relocation · Spanish (native) · English (B2)

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PROFESSIONAL SUMMARY

Software engineer with 6+ years of backend development experience, now focused on generative AI on AWS. Designed and shipped to production a RAG assistant built on Amazon Bedrock and pgvector, including its full infrastructure (ECS, RDS, VPC, ALB). Experienced in Python/Django, Node.js and .NET, IoT systems with edge-to-cloud synchronization, CI/CD, and mentoring developers.

TECHNICAL SKILLS

Generative AI: Amazon Bedrock (Agents, Titan Embeddings), RAG, embeddings, vector search with pgvector
AWS: ECS, ECR, RDS, S3, VPC, Application Load Balancer, Route 53, ACM, Bedrock
Backend: Python (Django), Node.js, C# / .NET Core, RESTful APIs, WebSockets, microservices
Databases: PostgreSQL, SQL Server (T-SQL, stored procedures), MySQL, SQLite
DevOps: Docker, Azure DevOps (CI/CD), Linux, Bash
Frontend: React, TypeScript, Angular
Methodology: Scrum, technical mentoring

PROFESSIONAL EXPERIENCE

AI Software Engineer — CTA Digital

Feb 2025 – Present

- Designed and built the backend and infrastructure for a generative AI (RAG) conversational assistant embedded in the company website, automatically answering customer questions about products; the company's first automated customer-support solution.
- Built the RAG pipeline over a ~500-document knowledge base: ingestion from Amazon S3, embeddings with Amazon Titan, vector search with pgvector on PostgreSQL (Amazon RDS), and answer generation with an Amazon Bedrock agent (Agent Runtime), exposed through a Django API.
- Designed and deployed the product's entire AWS infrastructure: containers on ECR and ECS behind an Application Load Balancer, private networking in a VPC, DNS with Route 53, and TLS certificates with ACM.
- Single-handedly developed the complete Smart Lockers (IoT) system: edge software in Python and Bash on Linux (IPC devices) that controls compartment locking and unlocking.
- Designed the edge-to-cloud sync architecture: a local Django backend on SQLite that synchronizes with PostgreSQL on Amazon RDS over WebSockets.
- Built a React (TypeScript) web dashboard showing the real-time status of each locker.

Stack: Python, Django, Amazon Bedrock (Agents, Titan), pgvector, PostgreSQL, SQLite, WebSockets, AWS (ECS, ECR, RDS, S3, VPC, ALB, Route 53, ACM), Docker, Linux, React, TypeScript

Programming Lead (Senior Software Engineer) — Grupo Ramos

May 2022 – Oct 2024

- Delivered enhancements to the Sirena Go e-commerce platform (thousands of users, ~30 stores): shopping cart, delivery orders, and product image gallery.
- Migrated legacy PHP services to a Node.js microservices architecture (3 services) with RESTful APIs and MySQL.
- Automated deployments with Azure DevOps CI/CD pipelines and secure secrets management for Docker-containerized applications, removing manual approval from every release.
- Mentored 4 developers and oversaw the team's code quality within a Scrum framework.
- Migrated the in-store self-service kiosk software to React (TypeScript).
- Built internal .NET (C#) tools, such as a barcode system for staff meal redemption.

Stack: Node.js, PHP, .NET (C#), MySQL, React, TypeScript, Docker, Azure DevOps, CI/CD

Programming Analyst & Backend Developer — Hunter del Caribe

Dec 2019 – May 2022

- Automated accounting and HR processes (bank reconciliation, payroll, daily sales closings, and social security (TSS) reporting) with SQL Server stored procedures, replacing manual Excel workflows.
- Maintained and extended the .NET Core backend and the company's internal web portals in Angular and React.

Stack: .NET Core, C#, SQL Server (T-SQL), Angular, React

EDUCATION

B.S. in Systems Engineering — Universidad O&M, Dominican Republic 2022

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-C03) — In progress · exam planned for January 2027

ADDITIONAL TRAINING

Docker Containers & Microservices — DevTalles · 2024 | Scrum Master Course — Eximedia Group · 2024